
GNN-BERT Fusion for Music Context Understanding

Abstract

This work asks whether a graph representation of audio structure, combined with a pretrained language model, improves music context understanding. Each audio clip becomes a *segment graph* encoded by GraphSAGE, and the accompanying text (tag metadata or a free-text caption) is encoded by DistilBERT. Four tasks are evaluated on MagnaTagATune (MTAT) and MusicCaps: text-only tagging, graph-only tagging, graph-text fusion with ablations, and contrastive audio-caption retrieval. On MTAT top-50 tagging, both fusion variants beat their unimodal ablations (early concatenation Macro-F1 0.278, cross-attention 0.272, against 0.217 for the text-only ablation and 0.242 for the graph-only ablation), and the contrastive model reaches Caption→Audio R@10 of 21.9%, roughly twelve times chance, with a zero-shot tag Macro-F1 of 0.164. Every model beats a prior-matched random baseline at 0.066. The analysis also reports where the design falls short: cross-attention buys nothing over concatenation, a band of rarer tags is never predicted above threshold at all, and the similarity-edge rule saturates its own degree cap on every MTAT graph. Code and reproducible artifacts accompany the report.

1 Introduction

Music understanding sits between audio signal processing and natural language. A piece has acoustic structure (repetition, contrast, texture) and describable semantics (genre, instrumentation, mood), and most tagging systems model only the first of these. Convolutional models on mel-spectrograms remain the standard approach on MagnaTagATune [2, 12], and they treat a clip as an undifferentiated image of time and frequency.

A segment graph makes a different assumption. Cutting a clip into overlapping windows and connecting windows that resemble each other turns repetition and return into an explicit edge rather than something a convolution has to rediscover from pixels. The question this project tests is whether that structure, once fused with a language model, adds anything a spectrogram model does not already provide.

The four tasks in the project specification are implemented end to end: text-only tagging, graph-only tagging, graph-text fusion with a four-way ablation, and contrastive audio-caption retrieval, each measured against three baselines on a fixed split with a fixed seed. Where a design choice could have gone either way, the report gives the measurement behind it, including two cases where the measurement showed the choice was doing nothing.

2 Related Work

Automatic music tagging on MTAT has been dominated by convolutional architectures on time-frequency inputs since [2], and [12] gives the comparison of those models under a shared folder split, which is the split convention this project follows. Graph neural networks enter music information retrieval mainly through symbolic representations such as chord or score graphs; applying

GraphSAGE [6] to audio-derived segment nodes, as here, trades spectral detail for an explicit encoding of self-similarity.

On the language side, the contrastive image–text objective of CLIP [9] transfers to audio through models such as CLAP [4], which pairs an audio encoder with a text encoder under InfoNCE [11]. Task 4 applies the same objective with a graph encoder in place of the audio encoder, at a data scale roughly three orders of magnitude smaller than CLAP’s, which is the context for the retrieval numbers reported later.

3 Method

Segment graphs. Each clip is resampled to 22.05 kHz mono and cut into 5 s windows with a 2.5 s hop. Each window becomes one node carrying a 64-dimensional vector: MFCC-20 and chroma-12 (extracted with librosa [8]), each summarised by mean and standard deviation over the window. Edges come in two kinds. Temporal edges connect adjacent windows in both directions. Similarity edges connect non-adjacent windows whose node features have cosine similarity above $\tau=0.6$, capped at 4 extra edges per node and weighted by the similarity value.

The 5 s window is long enough to contain a phrase or a bar group at common tempos, and short enough that a 29 s MTAT clip yields 11 nodes rather than 3 or 4; the 2.5 s hop gives each node a half-window overlap with its neighbour, so a musical event falling on a window boundary still lands inside some node. The value $\tau=0.6$ was chosen to keep only clearly related window pairs, and the degree cap of 4 was meant as a guard against dense graphs on repetitive material.

Measuring the built graphs shows the cap does all the work. Every MTAT graph has exactly 11 nodes, 20 temporal edges and 44 similarity edges, which is 4 per node on all 21,108 graphs. The cap binds everywhere, so the operative rule is “each window connects to its four most similar non-adjacent windows” and τ never rejects anything on this corpus. Raising τ would start to matter; at 0.6 it is inert. MusicCaps graphs have the opposite problem: a 10 s clip yields 3 nodes (occasionally 4), only one non-adjacent pair exists, and 99.5% of those graphs carry a similarity edge. Section 5 returns to what that does to the case-study evidence.

Encoders. The graph encoder is a 2-layer GraphSAGE [6] (PyTorch Geometric [5]) with mean readout, producing a 256-dimensional graph embedding g . Two layers give each node a receptive field of two hops, which on an 11-node graph already reaches most of the clip; deeper stacks over-smooth on graphs this small. Text is encoded by a fully fine-tuned DistilBERT [10, 3], keeping both the CLS embedding t_{CLS} and the token sequence H . DistilBERT rather than BERT-base because the 4 GB GPU used for every run has to hold the language model, the graph encoder and the fusion head at once.

Fusion (Task 3). The main fusion is single-head cross-attention with the graph readout as query and projected BERT tokens as keys and values,

$$\text{ctx} = \text{softmax}\left(\frac{qK^\top}{\sqrt{d}}\right)V, \quad q = gW_Q, \quad K, V = HW, \quad (1)$$

classified from $z = [g; \text{ctx}]$ through an MLP with sigmoid outputs under binary cross-entropy. Three ablations share the same head and training schedule: early concatenation ($z = [g; t_{\text{CLS}}]$), text-only ($z = t_{\text{CLS}}$), and graph-only ($z = g$). Holding the head and schedule fixed is what makes the four numbers comparable; the only thing that changes between them is which representation reaches the classifier.

Contrastive retrieval (Task 4). A dual encoder maps graphs and captions into a shared 256-dimensional L2-normalised space, trained with symmetric InfoNCE [11, 9] at temperature 0.07, with each branch warm-started from the Task-1 and Task-2 checkpoints. Warm-starting matters at this data scale: 4,817 training pairs are far too few to learn an audio encoder and a text encoder from scratch, so the contrastive stage aligns two encoders that already work rather than building them. Negatives are the other captions and graphs in the same mini-batch, with no memory bank and no hard-negative mining, so negative difficulty is bounded by the effective batch size of 64. This is the main structural gap against CLAP [4], which trains with far larger negative sets.

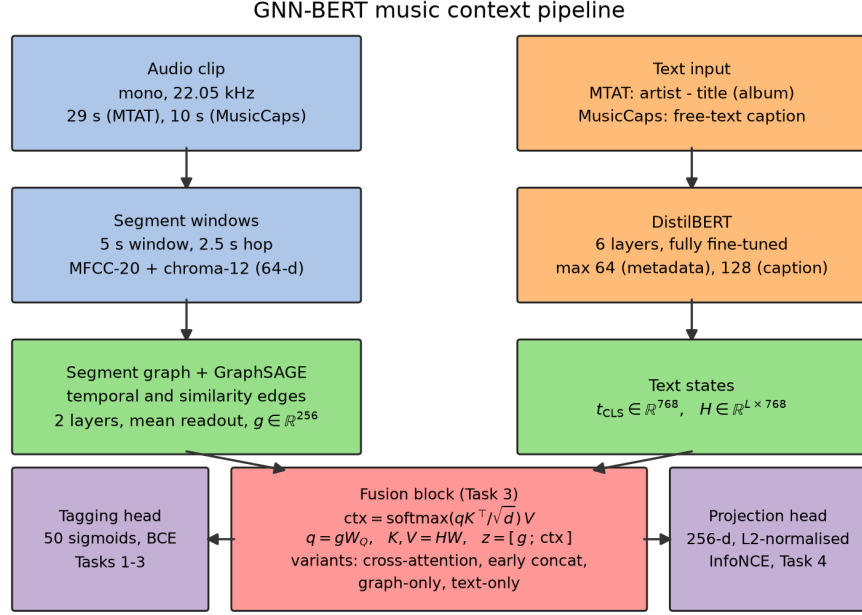


Figure 1: Pipeline. The left column turns a clip into a segment graph encoded by GraphSAGE; the right column encodes metadata or a caption with DistilBERT. The two meet in the fusion block, which feeds a 50-way sigmoid tagging head for Tasks 1–3 and a projection head trained with InfoNCE for Task 4. The four Task-3 variants are settings of the same fusion block, not separate paths.

Table 1: What restricting MTAT to its 50 most frequent tags costs. Counts are over the 25,863 annotated clips in the released annotation file.

Quantity	Value
Tags in the released vocabulary	188
Positive tag assignments in total	89,395
Assignments carried by the top 50 tags	69,714 (78.0%)
Assignments in the discarded 138-tag tail	19,681 (22.0%)
Clips with at least one tag	21,642
Clips that keep at least one top-50 tag	21,111
Clips that lose their entire label vector	531 (2.5%)
Tags occurring on fewer than 100 clips	67
Mean top-50 tags per retained clip	3.30

4 Experimental Setup

4.1 Datasets and label vocabulary

MagnaTagATune [7] supplies 25,863 annotated clips of 29 s each, labelled from a vocabulary of 188 tags. MTAT carries no lyrics or captions, so the Task-1 text input is the clip metadata string “artist - title (album)”. **MusicCaps** [1] supplies 10 s clips with expert free-text captions. YouTube blocks automated downloads of the original audio, so the audio comes from the public CLAPv2/MusicCaps mirror, giving 5,352 clips split 90/10 with seed 42. All runs use seed 42.

Following the project specification and the tagging literature [2, 12], MTAT is restricted to its 50 most frequent tags. That discards 138 of the 188 tags, so Table 1 sets out what the restriction keeps and what it throws away.

The discarded tail is largely redundant rather than novel. Fourteen of the dropped tags are spelling or inflection variants of retained ones: `electro` and `electronica` against `electronic`, `drum`

against drums, violins against violin, male vocals against male vocal, and harpsicord and clasical as outright misspellings. Those fourteen carry 3,646 of the discarded assignments, and on 68% of the clips holding them the canonical retained tag is already present, so dropping them removes nothing from those clips.

The tail is also sparse enough that supervised learning on it would measure noise. The median discarded tag appears on 100 of 25,863 clips, a prevalence of 0.39%. Under a fixed sigmoid threshold, a classifier for a tag that rare reaches near-perfect accuracy by always predicting zero, and its F1 is either 0 or a value decided by a handful of clips.

Macro-F1 makes that a problem for the headline number, because it weights every tag equally. Keeping all 188 tags would let 138 near-degenerate columns determine roughly three quarters of the reported score, which would then describe the annotation tail more than the model. Against that, the restriction costs 531 clips out of 21,642, which is a price worth paying.

The cut does lose genre coverage at the margins. jazz (439 clips), baroque (297) and folk (243) fall outside the top 50 despite naming genres no retained tag covers, so the label space skews toward instrument and mood descriptors. The t-SNE umbrellas in Section 5 inherit that skew.

4.2 Data splits

The 25,863 annotated MTAT clips are not 25,863 independent pieces of music. They are excerpts of 6,446 distinct recordings, averaging 4.87 excerpts per recording across the full clip table. A uniform random clip-level split would place, for a typical test excerpt, roughly four other excerpts of the same performance by the same artist into the training set, and a model could then score well by recognising the recording instead of reading the music.

MTAT ships its audio in 16 directories named 0–9 and a–f, and the directory prefix in each clip’s path is derived from artist, album and track, so a recording’s excerpts almost always share a directory. Splitting on directories therefore splits on recordings. Assigning 0–c to training, d to validation, and e–f to test gives 16,779 / 1,339 / 2,993 clips and leaves 11 of 6,446 recordings straddling the train–test boundary, with 3 straddling train–validation. That residue is small enough to ignore. This assignment puts one more directory in training than the 12/1/3 partition of [12], so the absolute numbers here are not directly comparable to that paper’s.

Artist-level leakage survives. Of the 56 artists appearing in the test directories, 36 also appear in training, covering 64% of test clips. MTAT draws from a catalogue of only 270 artists, so any split of it leaks artist identity unless artists are partitioned explicitly, which would shrink and unbalance the test set. Every number in Table 2 should be read as same-artist, unseen-recording performance rather than fully out-of-distribution performance.

4.3 Training

DistilBERT is fine-tuned at learning rate 2×10^{-5} while the graph encoder and fusion heads use 10^{-3} , a ten-times-larger rate for the randomly initialised parameters and a small one for the pretrained ones. Optimisation is AdamW with mixed precision and gradient accumulation, giving effective batches of 16–64. Epoch budgets are 10 for Task 1, 20 for Task 2, 15 for the CNN baseline, 15 for Task 3 and 20 for Task 4. Task 1 converges early (Figure 2, left panel, flat after epoch 5) which is why it gets the smallest budget. The checkpoint kept for test evaluation is always the one with the best validation Macro-F1, not the last epoch.

4.4 Baselines

Three baselines frame the results. B1 predicts each tag independently at its training prior, which admits a closed-form expected F1 and needs no training. B2 is a 4-block CNN on log-mel spectrograms, standing in for the conventional approach to this dataset [2]. B3 is the Task-1 BERT model, which isolates how much of the tagging score comes from metadata text alone.

4.5 Evaluation protocol

Tasks 1–3 report Macro-F1 and Micro-F1 at a sigmoid threshold of 0.5, fixed once and never tuned per model. Tuning a threshold per model would improve every number and destroy the comparison,

Table 2: MTAT test split, top-50 tags, threshold 0.5. Both fusion variants beat either unimodal ablation; every model beats the random baseline.

Model	Macro-F1	Micro-F1	mean AUC-PR
B1 random (tag priors)	0.066	0.098	0.066
B3 / Task 1 BERT (metadata)	0.189	0.318	0.302
Task 2 GraphSAGE	0.232	0.353	0.350
B2 CNN (log-mel)	0.290	0.427	0.400
Task 3 text-only ablation	0.217	0.333	0.294
Task 3 graph-only ablation	0.242	0.358	0.353
Task 3 early concatenation	0.278	0.400	0.378
Task 3 cross-attention	0.272	0.404	0.378

Table 3: Task 4 contrastive retrieval on the MusicCaps test split (535 clips) and zero-shot tag transfer to MTAT. A random ranker scores $R@10 \approx 1.9\%$.

Direction	R@1	R@5	R@10	Zero-shot tag probe	value
Caption $\rightarrow$ Audio	5.23	14.58	21.87	Macro-F1	0.164
Audio $\rightarrow$ Caption	4.11	13.27	20.75	Micro-F1	0.179
				mean AUC-PR	0.162

since a model with poorly calibrated probabilities could overtake a better one by finding a friendlier operating point. Mean AUC-PR is reported alongside because it is threshold-free and therefore immune to that concern.

The Task-4 zero-shot probe cannot use the same rule. It ranks tags by cosine similarity between a clip embedding and tag-prompt embeddings, and a cosine similarity is not calibrated to $[0, 1]$ the way a sigmoid output is; a fixed cut at 0.5 would predict nothing at all. The similarity threshold is therefore swept for best F1 on a 500-clip MTAT validation subset and the resulting value 0.151 is applied unchanged to the test subset. Because the selection procedure differs, the zero-shot Macro-F1 of 0.164 is not comparable to Table 2 as a tagging accuracy, only as evidence that the contrastive space transfers.

5 Results

Tagging (Table 2). Audio structure alone (GraphSAGE, 0.232) beats sparse metadata text (BERT, 0.189). The raw-spectrogram CNN is the strongest single model at 0.290, which follows from what each input preserves: a 64-dimensional MFCC and chroma summary per 5 s window throws away the fine spectral detail a convolution reads directly off the mel-spectrogram.

Fusing the two modalities beats either one alone: early concatenation reaches 0.278 and cross-attention 0.272, against 0.217 text-only and 0.242 graph-only, with all four sharing the same head and schedule. The gain over the better single modality is 0.036 Macro-F1, about a 15% relative improvement.

The two fusion variants are tied within noise. Early concatenation leads on Macro-F1 by 0.006, cross-attention leads on Micro-F1 (0.404 against 0.400), and the two match on AUC-PR at 0.378, so no advantage is claimed for either. The likely reason cross-attention buys nothing is the text itself: the MTAT metadata string averages a handful of content words, the CLS summary already carries most of what those words say, and attention over four or five informative tokens has little structure left to exploit. Cross-attention should be expected to earn its extra parameters on caption-length text, not on “artist - title (album)”.

The Macro–Micro gap. Every model in Table 2 scores far lower on Macro-F1 than on Micro-F1, and Figure 3 shows where the difference comes from. Per-tag F1 rises with prevalence, and a band of tags sits at exactly zero: the model never assigns them a probability above 0.5 on any test clip, so their F1 contribution is zero regardless of how well the model ranks them. Average precision for

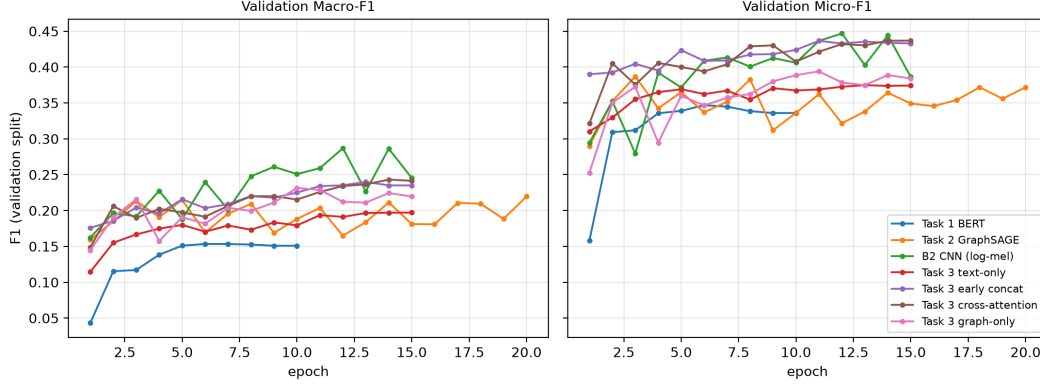


Figure 2: Validation Macro-F1 (left) and Micro-F1 (right) against epoch. One colour per model, shared across panels. Task 1 flattens after epoch 5; the Task-3 variants separate on Macro-F1 from about epoch 8 onward, while the CNN baseline is noticeably noisier between epochs than any of them.

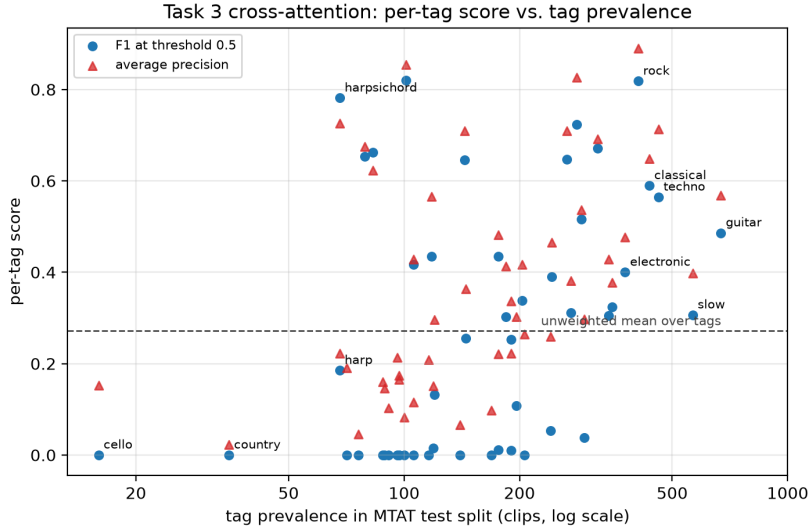


Figure 3: Per-tag F1 and average precision against tag prevalence in the MTAT test split, log-scaled x-axis. Tags sitting on the zero line are never predicted above threshold; their average precision shows the ranking is not uninformative.

those same tags is generally well above zero, which says the ranking is informative and the fixed threshold is what discards it. Micro-F1 hides this because it pools decisions globally and is therefore dominated by the few tags that appear on hundreds of clips; Macro-F1 gives cello the same weight as guitar and reports the damage.

This is the direct consequence of the protocol choice in Section 4.5. A per-tag threshold would raise Macro-F1 substantially, and it would also make the cross-model comparison meaningless. The fixed threshold is the honest setting and the low Macro-F1 numbers are its price.

Retrieval (Table 3). The dual encoder reaches $R@10 \approx 22\%$ on caption-to-audio retrieval over 535 candidates, roughly twelve times the 1.9% a random ranker would achieve, and it transfers zero-shot to MTAT tagging at Macro-F1 0.164 using only tag-name prompts, having never seen an MTAT tag during contrastive training. That transfer says the shared space encodes something about tag semantics rather than only the MusicCaps caption distribution.

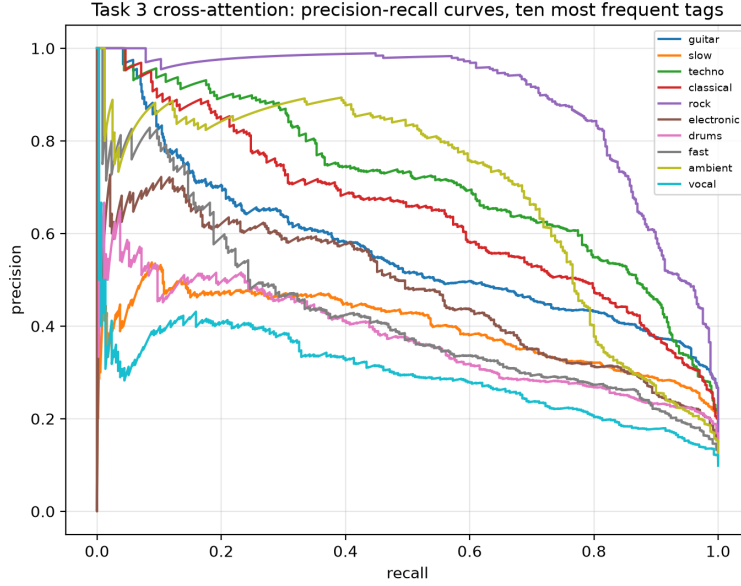


Figure 4: Precision–recall curves for the ten most frequent MTAT tags under the cross-attention model. rock and techno hold high precision deep into recall, while vocal and slow degrade almost immediately, which matches their status as broad descriptors that apply across genres.

Analysis of the embedding space. Figure 5 projects the fused Task-3 embedding of the MTAT test split and colours it by genre umbrella. Vocal clips gather at the top left, classical occupies a band beneath them along with several tight clusters through the centre-left, ambient runs from the mid-left down into the bottom centre, electronic fills the centre right and lower right, and rock forms a compact island at the far right. Acoustic, the largest umbrella, threads between the others rather than claiming a region, which is what an umbrella built from guitar, country and solo should do. The frequent umbrellas separate; world (38 test clips), instrumental (19) and pop (6) are too small to form regions and appear as scattered points.

Figure 6 repeats the projection coloured by mood umbrella. Calm (1,007 clips) and energetic (1,554) occupy broadly opposite sides of the map, while the smaller dark (132) and upbeat (7) classes stay mixed into the calm side. Broad arousal organises the space; finer affective distinctions do not. Part of that is the umbrella mapping itself, which assigns opera and sitar to “dark” on timbral grounds that a listener might well dispute, so the weak separation of the small classes is partly a labelling artifact rather than a pure model failure.

Qualitative behaviour and failure cases. On a Bach cantata clip the demo predicts {opera, classical, violin, strings}, an exact match to its four ground-truth tags. Caption-to-audio retrieval for a soft female-vocal pop query returns clips sharing the soft female-vocal character (a lightly sad female vocal, a lullaby) but not the pop genre, which is the pattern across the sampled queries: the contrastive space captures timbre and mood more reliably than genre.

The three case studies of well-retrieved clips are all 3-node graphs carrying 2 similarity edges on top of the 4 temporal ones, and it is tempting to read that as evidence that self-similar structure aids alignment. The corpus statistics refute it. A 10 s MusicCaps clip yields three 5 s windows, exactly one non-adjacent pair exists, and 99.5% of MusicCaps graphs carry that edge. The case studies therefore show nothing the average clip does not also show, and the self-similarity hypothesis cannot be tested at this graph size at all. Testing it would need longer source clips.

The text branch’s failure mode is visible on clip 22199 (Briddes Rounne - Edi be thu (Lenten is come), true tags flute, harp), where the model predicts {female, woman, vocal, female vocal, opera}. The album title “Lenten is come” also names clip 2406, a genuinely vocal piece the model tags correctly, so the prediction is driven by the album string rather than by anything de-

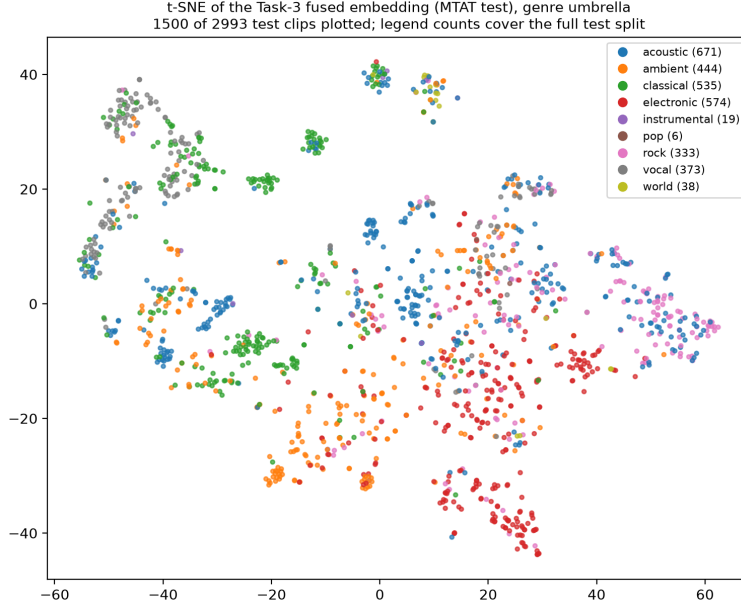


Figure 5: t-SNE of the Task-3 fused embedding z on the MTAT test split, coloured by genre umbrella.

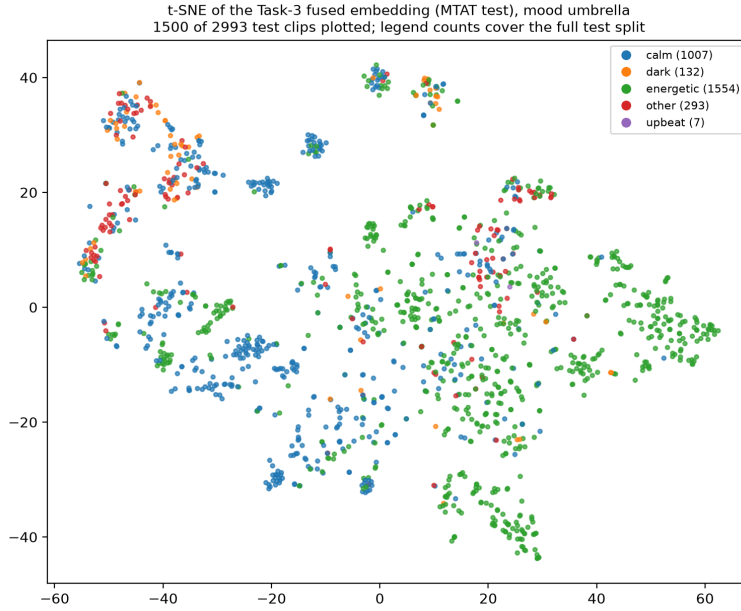


Figure 6: The same projection coloured by mood umbrella.

scribing this clip’s instrumentation. Metadata text carries album-level information, and album-level information is wrong at clip level whenever an album is not uniform.

Across ten sampled retrieval queries, only one returns the literal ground-truth clip inside the top 3 (rank 2, cosine similarity 0.483), which is what a caption-to-audio $R@1$ of about 5% predicts. The rest return semantically related but different clips; a scratching query, for instance, returns three separate turntablism clips. Aggregate $R@10$ of 22% is therefore driven by broad topical retrieval rather than by frequent exact matches, and reading it as “the model finds the right clip one time in five” would overstate it.

6 Limitations

Node features are low-dimensional hand-crafted descriptors, which caps the graph branch below the CNN baseline; learned per-segment audio embeddings are the obvious way to close that gap and the clearest single change to make next.

Task-1 text is metadata rather than lyrics or captions, so the language branch is weak on MTAT by construction, and the clip-22199 failure above shows the specific mechanism.

The similarity-edge rule never exercises its threshold on MTAT and has almost no room to act at all on MusicCaps, so the ablation that would matter most, temporal edges against temporal-plus-similarity edges, was never run.

Artist-level leakage survives the folder split, as Section 4.2 quantifies, so the tagging numbers describe unseen-recording rather than unseen-artist generalisation.

Contrastive training sees at most 63 negatives per anchor because negatives come only from the mini-batch. CLAP-scale retrieval numbers are not reachable under that constraint on a 4 GB GPU.

Human evaluation covers 10 caption–clip pairs rated 1–5 by a single rater, scoring mean 2.75 with standard deviation 1.4. One rater on ten pairs supports no statistical claim; the specification asks for at least five listeners and this project did not reach that. Two of the ten sampled captions independently retrieved the same top-1 clip, which is consistent with an R@1 of about 5%: at that confidence level top-1 collisions across unrelated queries are expected rather than a labelling error.

7 Conclusion

Fusing an audio segment-graph encoder with a fine-tuned language model beats either modality alone on multi-label tagging, by 0.036 Macro-F1 over the better single modality, and the same architecture supports cross-modal retrieval and zero-shot tag transfer. Cross-attention does not beat plain concatenation on metadata-length text, so the extra machinery is not justified at this text length. The segment graph runs on hand-crafted node features; replacing them with learned per-segment audio embeddings, and testing the similarity edges on clips long enough for the threshold to bite, are the two changes most likely to move the numbers.

References

- [1] Andrea Agostinelli, Timo I Denk, Zalán Borsos, Jesse Engel, et al. MusicLM: Generating music from text. *arXiv preprint arXiv:2301.11325*, 2023. Introduces the MusicCaps dataset.
- [2] Keunwoo Choi, György Fazekas, and Mark Sandler. Automatic tagging using deep convolutional neural networks. In *ISMIR*, 2016.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *NAACL-HLT*, 2019.
- [4] Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. CLAP: Learning audio concepts from natural language supervision. In *ICASSP*, 2023.
- [5] Matthias Fey and Jan Eric Lenssen. Fast graph representation learning with PyTorch Geometric. *arXiv preprint arXiv:1903.02428*, 2019.
- [6] Will Hamilton, Zhitao Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *NeurIPS*, 2017.
- [7] Edith Law, Kris West, Michael I Mandel, Mert Bay, and J Stephen Downie. Evaluation of algorithms using games: The case of music tagging. In *ISMIR*, pages 387–392, 2009.
- [8] Brian McFee, Colin Raffel, Dawen Liang, Daniel PW Ellis, et al. librosa: Audio and music signal analysis in python. In *SciPy*, 2015.
- [9] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

- [10] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- [11] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- [12] Minz Won, Andres Ferraro, Dmitry Bogdanov, and Xavier Serra. Evaluation of CNN-based automatic music tagging models. In *Sound and Music Computing Conference (SMC)*, 2020.